SAI YASASWINI MAJETY

Senior Full Stack Developer | Enterprise Cloud Platforms & Distributed Systems

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Professional Summary

- **Over 6+ Years of Production Full Stack & Backend Engineering Experience:** Senior Full Stack Developer with deep enterprise experience architecting, scaling, and hardening mission-critical distributed systems, full stack web applications, and regulated cloud platforms across commercial aviation and advanced AI research.
- **Enterprise Java & Spring Boot Core Platform Engineering:** Extensive production expertise developing robust microservices in **Java 21, Java 17, and Spring Boot 3.2**, utilizing Spring Data JPA, Hibernate, Spring Security, HikariCP, and Maven to build resilient transactional APIs handling multi-million daily requests.
- **Modern Frontend Architecture (React 18 & TypeScript):** Hands-on mastery building responsive, accessible, single-page enterprise applications in **React 18.3, TypeScript 5.4**, and Next.js; expert in component lifecycle design, RouteGuard role-based access control, state management (Redux Toolkit, React Query), and real-time WebSockets.
- **Regulated Data Governance & Audit Traceability:** Track record engineering high-assurance architectures meeting strict regulatory mandates; deep expertise implementing dual-layer immutable audit trails (trigger-based mutation capture and structured JSON event streaming) ensuring tamper-evident data provenance, non-repudiation, and rigorous compliance reporting.
- **Distributed Orchestration, Workflow Engines & Messaging:** Deep expertise orchestrating complex multi-step approval workflows using **Temporal SDK**, event-driven microservices with **Apache Kafka** and AWS SQS/SNS, Saga pattern compensation transactions, soft inventory reservations with TTL expiry, and Redis caching.
- **Zero-Downtime Relational Database Migrations:** Specialized in high-availability **PostgreSQL 16** and AWS RDS Aurora Multi-AZ; author of idempotent, versioned **Flyway DDL migrations**, table partitioning, optimistic locking, and shadow table cutovers executing with <50ms locking.
- **Enterprise Cloud Infrastructure & Container Orchestration:** Production engineering across AWS (ECS Fargate, RDS, S3, IAM, CloudWatch) and Red Hat OpenShift / Kubernetes; authoring declarative Terraform IaC and Helm charts with automated CI/CD deployment pipelines.
- **Enterprise Identity, Security & Zero-Trust Governance:** Deep practical knowledge implementing OAuth 2.0 / SAML 2.0 federation, Azure AD / Okta MSAL identity gateways, automated TLS/SSL X.509 certificate rotations, and least-privilege IAM controls.
- **Executive Recognition & High-Impact Delivery:** Awarded **2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg, 2 internal merit promotions, and 1st Place winner in the Viasat Global IoT Hackathon across 24 international engineering teams.

Technical Skills Matrix

Enterprise Backend Frameworks: Java 21 / 17 / 11, Spring Boot 3.2/3.x, Spring Data JPA, Spring Security 6, Hibernate ORM, Python (FastAPI, Django, Flask), Go, RESTful APIs, gRPC, HikariCP, Maven, uvloop

Modern Frontend Technologies: React.js 18.3, TypeScript 5.4, Angular, Next.js, TanStack Query (React Query), Zod, Jotai Atoms, Redux Toolkit, Material UI (MUI), Design Tokens / Style Dictionary, Vite, RouteGuard Security

Workflow Orchestration & Messaging: Temporal SDK (Orchestration Workflows, Activity Retries), Apache Kafka, RabbitMQ, AWS SQS, AWS SNS, Redis (ElastiCache, Pub/Sub, In-Memory Caching), Event-Driven Architecture, CQRS

Databases & Schema Migrations: PostgreSQL 16, AWS RDS Aurora Multi-AZ, MySQL 8.0, Flyway Idempotent Schema Migrations, JPA/Hibernate Entity Reconciliation, Custom ID Generators, Optimistic Locking, Soft Reservation TTLs

Cloud Infrastructure & Containers: AWS (ECS, RDS Aurora, S3, IAM, CloudWatch, KMS, STS, Secrets Manager), Red Hat OpenShift, Docker, Kubernetes, Helm, Terraform (IaC), ArgoCD, Jenkins Declarative Pipelines

Security, Identity & Compliance: OAuth 2.0 / SAML 2.0 Federation, Azure AD / Okta MSAL, Zero-Trust Security Filters, Automated TLS/SSL X.509 Rotation, Immutable Audit Trails, High-Assurance Data Governance

Testing, Quality & Observability: JUnit 5, Mockito, AssertJ, PyTest, Playwright, Vitest, Jest, React Testing Library, SonarQube, JaCoCo Coverage Enforcement, Spotless Code Formatting, OpenTelemetry, Splunk, Datadog

Architecture, Integration & Tooling: Microservices, Asynchronous Request Pipelines, Saga Compensation Patterns, Optimistic Mutation Audits, REST/JSON, Git, Jira, Confluence, Linux/Unix Shell Scripting, Swagger/OpenAPI

Flagship Honors, Awards & Technical Credentials

- **Viasat CEO Performance Awards (2x Recipient):** Commended directly by CEO Mark Dankberg for five-nines (99.999%) platform availability and zero-downtime fleet-wide operational reliability.
- **Viasat Spotlight Recognition Award:** Awarded for enterprise security innovation in automating PKI TLS/SSL X.509 certificate lifecycle and zero-touch cryptographic rotation across 150+ edge endpoints.
- **Viasat Architecture Commendation:** Formally recognized for engineering decoupled OAuth 2.0 / SAML identity gateway microservices that reduced partner enterprise onboarding time by **90%**.
- **1st Place Winner, Viasat Global IoT Hackathon (1st of 24 teams):** Architected and prototyped an edge telemetry diagnostic ingestion pipeline with real-time React dashboarding in 24 hours.
- **Dean's Merit Scholarship Recipient (SASTRA University):** Awarded for sustained academic distinction, ranking in the top 3% of the cohort throughout undergraduate engineering tenure.

Professional Experience

Viasat Inc. : Commercial Aviation Platform & Security Infrastructure **Chennai, India / Carlsbad, CA**

Senior Software Engineer - I (Full Stack & Cloud Platform Engineering) *Jan 2023 - Feb 2024*

Environment: Java 21, Spring Boot 3.2, React 18, TypeScript, Python, PostgreSQL 16, AWS RDS Multi-AZ, Kafka, Redis, Docker, Kubernetes, OpenShift, Flyway, OAuth 2.0, Okta, JUnit 5, Mockito

- **Sustained a strict five-nines (99.999%) availability SLA** across 1,000+ commercial aircraft (Delta Air Lines fleet) by architecting resilient backend microservices in **Java 21, Spring Boot 3.2**, and Python with non-blocking asynchronous request routing, Redis caching, and circuit-breaker fault tolerance.
- **Built responsive real-time operational web dashboards in React 18, TypeScript, and Angular** with Jotai reactive state, isolating multi-step form workflows across concurrent browser sessions to eliminate state collision and session draft loss.
- **Eliminated 100% of customer-facing database downtime** across multi-terabyte transactional clusters on AWS RDS PostgreSQL 16 by executing zero-downtime schema refactoring with idempotent **Flyway DDL scripts**, optimized JPA child collection reconciliation, and high-performance sequence ID generation.
- **Automated enterprise PKI certificate lifecycle across 150+ edge endpoints**, authoring zero-touch X.509 rotation daemons and synthetic expiration monitoring hooks that eliminated 100% of certificate outage risks (**Viasat Spotlight Recognition Award**).
- **Accelerated enterprise partner identity onboarding by 90%** (from 3 weeks to 2 days) by architecting decoupled **OAuth 2.0 and SAML/Okta identity gateway microservices**, enforcing zero-trust Spring Security JWT filter chains and backend **@PreAuthorize** access control.
- **Achieved 100% immutable audit traceability** across microservice state transitions by implementing mutation change-detection algorithms and dual-layer audit logging (database trigger mutation capture via PostgreSQL JSON functions paired with structured event streaming), satisfying strict compliance and security audit mandates.
- **Reduced session synchronization failures to <0.01%** across intermittent satellite uplinks by engineering distributed message queues with idempotent consumer processing, saga compensation transactions, and exponential backoff retry policies.
- **Decoupled high-velocity telemetry ingestion** from downstream analytical databases by implementing Apache Kafka event streams, eliminating database write contention and sustaining 12,000+ events/second during peak commercial flight banks.
- **Awarded 2 CEO Performance Bonuses** directly by Viasat Chairman & CEO Mark Dankberg for exceptional architectural execution and operational stability during major airline fleet software rollouts.

Viasat Inc. : Cloud Platform & API Microservices **Chennai, India**

Software Engineer II (Full Stack Platform & Microservices Architecture) *Jan 2021 - Dec 2022*

Environment: Java 17, Spring Boot, React, TypeScript, Python, AWS (ECS, RDS, S3, IAM, CloudFormation), Docker, OpenShift, Jenkins CI/CD, PyTest, JUnit 5, RESTful APIs, Git, Linux

- **Accelerated multi-region cloud deployment velocity by 65%** (cutting MTTR from 4 hours to 45 minutes) by authoring modular AWS CloudFormation IaC templates and declarative Jenkins CI/CD pipelines across staging and production clusters.
- **Elevated automated test coverage from 60% to >92%** across critical authentication, billing, and fleet session services by architecting comprehensive JUnit 5, Mockito, and PyTest test suites with automated network failure simulations.
- **Engineered automated cross-region database replication monitoring** and health failover probes for enterprise AWS RDS instances, guaranteeing transactional recovery objectives (RPO < 1 minute, RTO < 15 minutes) during disaster recovery drills.
- **Developed full stack administrative portal features** in React, TypeScript, and Java Spring Boot, enabling tier-2 customer support and ground operations teams to manage aircraft access permissions and inspect real-time billing logs.
- **Cut container build and deployment pull times by 50%** (from 14 minutes to 7 minutes) by re-architecting multi-stage Docker builds and layer caching strategies across Kubernetes and OpenShift deployment manifests.
- **Enforced strict API contract governance** by authoring OpenAPI/Swagger 3.0 interface definitions and implementing contract validation filters, eliminating API drift between frontend React components and backend microservices.
- **Honored with the 2021 Viasat CEO Mark Dankberg Performance Bonus** for high-impact infrastructure delivery, zero defect escapes, and outstanding operational discipline.

Viasat Inc. : Distributed Systems, Search & Tooling Automation **Chennai, India**

Software Engineer I & Software Engineer Intern (Full Stack & Tooling) *Jan 2018 - Dec 2020*

Environment: Java 11, Python 3.8, Django, Flask, React, JavaScript, PostgreSQL, MariaDB, Sphinx Search, Docker, Jenkins, Git, Linux, Robot Framework, JUnit

- **Commercial Air PAT (Pipeline Observability & Quality Dashboard):** Engineered an enterprise observability platform in Python Flask and React aggregating build results and defect telemetry across **50+ Jenkins nodes**, cutting release verification latency by **40%**.
- **CandEx 2.0 Platform (Internal Operations & Search Engine):** Accelerated search query response times from **3.2s to <180ms** (17x speedup) by integrating Sphinx full-text search with Python Django microservices across 10,000+ relational database records.
- **Automated pre-merge regression verification pipelines** in Groovy and Robot Framework, executing 100+ automated test scenarios per release candidate and preventing breaking API regressions from reaching staging environments.
- **Engineered automated ETL data extraction pipelines** that normalized heterogeneous sensor logs into PostgreSQL, accelerating daily operational reporting generation from **4 hours to 15 minutes**.
- **Built responsive web telemetry interfaces in React** displaying real-time terminal connection states, RF link signal-to-noise ratios, and historical uptime trends for flight network operations centers.
- **Won 1st Place out of 24 international engineering teams in the Viasat Global IoT Hackathon** for designing an end-to-end connected IoT sensor prototype with cloud telemetry ingestion and live web dashboarding in 24 hours.

Professional Experience (Continued)

Independent AI Systems & Full Stack Cloud Platform Research — **Houston, TX**

Lead Systems & Cloud Infrastructure Researcher — *Mar 2024 - Present*

Environment: Java 21, Spring Boot 3.2, Python 3.12, React 18, TypeScript, PyTorch, FastAPI, FAISS, Docker, AWS (ECS Fargate, S3, RDS), Terraform, GitHub Actions, Hugging Face

- **Architected asynchronous microservice APIs & distributed caching proxies** using Java 21, Spring Boot 3.2, and Redis, achieving **sub-15ms P99 latency** while sustaining **12,000+ req/sec** under simulated enterprise peak loads.
- **Engineered high-throughput Retrieval-Augmented Generation (RAG) backend services** using Python FastAPI, FAISS vector indexing, and transformer embeddings, achieving **sub-200ms semantic retrieval** over complex technical documentation.
- **Architected containerized microservices deployments on AWS** (ECS Fargate, RDS PostgreSQL, S3, IAM) utilizing Terraform Infrastructure-as-Code and automated GitHub Actions CI/CD workflows, cutting inference overhead by **35%**.
- **Developed full stack exploratory portals in React 18 and TypeScript** for interactive model evaluation, providing researchers with dynamic vector similarity visualization, token attribution heatmaps, and latency benchmarking dashboards.
- **Engineered end-to-end integration and UI validation suites** using Playwright and Jest, validating interactive component state rendering, API error recovery states, and asynchronous server-sent event (SSE) streams.

Flagship Systems Architectures & Technical Projects

Clinical Logistics & Workflow Orchestration Platform — **Java 21** | **Spring Boot 3.2** | **React 18** | **Temporal** | **PostgreSQL 16**

- Architected an enterprise clinical supply and inventory orchestration platform modeled on time-sensitive perishable healthcare logistics, implementing a dual-constraint reservation engine balancing perishable product shelf-life decay curves alongside physical batch fill limits.
- Engineered multi-segment cold-chain distribution workflows with intermediate regional distribution hubs, integrating automated temperature logging validations and multi-step Temporal SDK approval state machines distinguishing clinical protocol verification from automated fulfillment dispatch.
- Enforced zero-drift schema governance across 40+ relational tables using Flyway idempotent migrations, single-table inheritance with discriminator columns, and immutable database audit triggers capturing old/new JSON entity snapshots with complete historical traceability.

High-Throughput Semantic Vector Caching Microservice — **FastAPI** | **Redis** | **Pydantic v2** | **Docker** | **AWS**

- Engineered a sub-20ms semantic caching proxy in FastAPI that intercepts API queries, computes embedding hashes, and performs cosine distance matching against Redis in-memory vector indices, achieving an 85% cache hit rate under 12,000+ req/sec.
- Optimized asynchronous request pipelines using compiled **uvloop** event loops and Pydantic v2 binary serializations, slashing downstream relational database query loads by **60%**.
- Built an operational health monitoring UI in React and Chart.js, rendering real-time cache hit ratios, memory eviction rates, and downstream p99 latency distributions for platform engineers.

Open-Source Enterprise RAG Assistant (rag-hf-chatbot) — **Python 3.11** | **Hugging Face** | **FAISS** | **Streamlit** | **Docker**

- Engineered an interactive document question-answering system using Hugging Face transformer embeddings and FAISS vector indexing, achieving sub-200ms semantic context retrieval over complex technical documentation with hybrid BM25 + dense ranking.
- Implemented conversational context memory management and structured source citation grounding, eliminating model hallucination across multi-turn user dialogues with automated verification checks.
- Designed multi-tenant data isolation filters and session token encryption, preventing cross-tenant document visibility and ensuring enterprise-grade data security across shared runtime containers.

Edge Store-and-Forward Telemetry Daemon — **Python 3.11** | **SQLite WAL** | **Zstandard** | **Embedded Linux**

- Built an asynchronous local queueing engine for intermittent networks using SQLite WAL ring buffering, guaranteeing zero telemetry loss during link dropouts while cutting uplink bandwidth consumption by **65%** via Zstandard streaming compression.
- Engineered crash-resilient transactional recovery routines in Python and C++, preventing database corruption and ensuring zero-packet loss during sudden inflight power cycles.

Technical Certifications & Professional Training

- **Enterprise Systems & Industrial Automation Training (HCL Technologies):** Enterprise IT infrastructure, automation hardware, telemetry communication protocols, and industrial monitoring standards.
- **Microcontroller & Embedded Systems Certification (ECIL):** Embedded C programming, telemetry hardware interfacing, PIC microcontroller architecture, and serial communication protocols.

Education & Academic Credentials

SASTRA Deemed University — **Thanjavur, India**

Bachelor of Technology (B.Tech) in Electrical and Electronics Engineering (EEE) — *Aug 2014 - May 2018*

Academic Distinction: Cumulative GPA: **8.2 / 10.0** | Recipient of **Dean's Merit Scholarship** (Top 3% cohort distinction).
Core Coursework: Object-Oriented Programming (Java, C++), Data Structures & Algorithms, Relational Databases & SQL, Distributed Systems, Computer Networks, Operating Systems; Capstone: Embedded Sensor Telemetry System.